

Subject Description Form

Subject Code	DSAI4208
Subject Title	Introduction to Machine Vision and Intelligence
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	• Introduce students to the basic concepts and principles of computer vision and artificial intelligence. • Familiarize students with key computer vision and machine learning algorithms and techniques. • Develop students' ability to apply machine vision and intelligence concepts to solve practical problems.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: a) Describe the core principles of computer vision and artificial intelligence. b) Utilize essential computer vision algorithms for solving visual recognition tasks. c) Apply computer vision and deep learning methods to address practical challenges in various fields. d) Show foundational ability to integrate machine learning techniques with computer vision for developing basic intelligent systems.
Subject Synopsis/ Indicative Syllabus	1. Image Acquisition and Representation • Basics of image representation and video representation • Introduction to color fundamentals (RGB, CMYK, HSL/HSV) 2. Image Processing Techniques • Introduction to image filters and basic convolutional filters • Edge detection (image gradient, Sobel filter) • Basic image matching concepts 3. Machine Learning for Vision • Introduction to supervised and unsupervised learning • Introduction to neural networks and deep learning • Understanding model generalization in machine learning 4. Visual Recognition Tasks and Algorithms • Image classification • Object detection and segmentation • Fundamentals of image understanding and generation 5. Applications • Introduction to 3D vision and its role in robotics. • Everyday applications of machine vision and intelligence, such as facial recognition, medical imaging, autonomous vehicles.

Teaching/Learning Methodology	The course will be delivered through a mix of lectures and tutorials, providing students with a solid understanding of the fundamental concepts and technologies involved in visual data processing, representation, and applications.					
Assessment Methods in Alignment with Intended Learning Outcomes	Specific Assessment Methods/Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed			
			a	b	c	d
	1. Assignments	20%	✓	✓	✓	
	2. Projects	20%		✓	✓	✓
	3.. Final Examination	60%	✓	✓	✓	
	Total	100 %				
	Examinations and assignments are designed to assess students’ understanding of machine vision concepts and algorithms. The project, meanwhile, evaluates students’ practical skills in applying machine learning methods and AI models to process visual data and solve real-world problems.					
Student Study Effort Expected	Class contact:					
	• Lectures / Tutorials / Labs					39 Hrs.
	Other student study efforts:					
	• Assignments and Projects					36 Hrs.
	• Self-study					30 Hrs.
	Total student study effort:					105 Hrs.
Reading List and References	References: 1. R. Shanmugamani, <i>Deep Learning for Computer Vision</i> , Packt Publishing, Jan 2018. 2. Zhang, Aston and Lipton, Zachary C. and Li, Mu and Smola, Alexander J., <i>Dive into Deep Learning</i> , arXiv preprint arXiv:2106.11342, 2021. Website: https://d2l.ai/ . 3. Mohamed Elgendy, <i>Deep Learning for Vision Systems</i> , October 2020. 4. Richard Szeliski, <i>Computer Vision: Algorithms and Applications</i> , Springer, 2022. Free for download at https://szeliski.org/Book/ . 5. Computer Vision: Models, Learning, and Inference http://www.computervisionmodels.com/ 6. Computer Vision Lectures: https://fpcv.cs.columbia.edu/ (also available on YouTube) 7. François Chollet, <i>Deep Learning with Python</i> . https://sourestdeeds.github.io/pdf/Deep%20Learning%20with%20Python.pdf 8. Keith Price Bibliography Annotated Computer Vision Bibliography: Table of Contents					